Index

Unit – 3 ⇝ Linear Transformation (Tutorial Manual)..... 3

1) Method 1 ⇝ Linear Transformations 3
2) Method 2 ⇝ Kernel, Range and Rank-Nullity Theorem 5
3) Method 3 ⇝ Matrix Representation of Linear Transformations.....9
4) Method 4 ⇝ Change of Basis.....13
5) Method 5 ⇝ Similarity of Matrices.....17

Unit – 3 ⇝ Linear Transformation (Tutorial Manual)

Method 1 ⇝ Linear Transformations

Examples of Method-1: Linear Transformations

A	1	Check whether $T : \mathbb{R}^3 \to \mathbb{R}$ defined by $T(x_1, x_2, x_3) = 2x_1 - 3x_2 + 4x_3$ is linear transformation or not? **Answer: Yes**
A	2	Check whether the transformation $T : \mathbb{R}^2 \to \mathbb{R}$ defined by $T(x, y) = \mid 2x - 3y \mid$ is linear transformation or not? **Answer: No**
A	3	Check $T : \mathbb{R}^2 \to \mathbb{R}^3$, where $T(x_1, y_1) = (x_1 + y_1,\ x_1 - y_1,\ y_1)$ is linear transformation or not? **Answer: Yes**
A	4	Check whether the function $T : \mathbb{R}^2 \to \mathbb{R}^2$ given by the formula $T(x,\ y) = (x + 2y,\ 3x - y + 5)$ is linear transformation or not? **Answer: No**
B	5	Check whether the transformation $T : \mathbb{M}_{m\times n} \to \mathbb{M}_{n\times m}$, where $T(A) = A^T$ is linear transformation or not? **Answer: Yes**
B	6	Check whether the following transformation are linear or not? (i) $T : \mathbb{M}_{n\times n} \to \mathbb{R}$, where $T(A) = \det(A)$. (ii) $T : \mathbb{M}_{n\times n} \to \mathbb{R}$, where $T(A) = \mathrm{tr}(A)$. **Answer: (i) No, (ii) Yes**
B	7	Check whether the following transformation are linear or not? (i) $T : \mathbb{P}_1 \to \mathbb{P}_2$, where $T\big(p(x)\big) = x\,p(x) + x^2 + 1$. (ii) $T : \mathbb{P}_1 \to \mathbb{P}_2$, where $T\big(p(x)\big) = x\,p(x) + p(0)$. **Answer: (i) No, (ii) Yes**

C	8	Define a transformation $T : \mathbb{P}_3 \to \mathbb{P}_2$ by $T(p(x)) = p'(x) + p''(x)$, where $p'(x)$ and $p''(x)$ denote the first and second derivatives of $p(x)$, respectively. Show that T is a linear transformation. Find the image of the polynomial $q(x) = 2x^3 - 3x^2 + 4x - 1$. **Answer: Yes, $T(q(x)) = 6x^2 + 6x - 2$**
B	9	Define a transformation $T : \mathbb{P}_2 \to \mathbb{P}_3$ by $T(p(x)) = (x + 1)\,p(x)$. Check whether T is a linear transformation or not. Also find the image of the polynomial $p(x) = x^2 - 2x + 3$. **Answer: Yes, $T(q(x)) = x^3 - x^2 + x + 3$**
C	10	Define function $T : \mathbb{R}^3 \to \mathbb{R}^2$ by $T\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$. Find the image of set $S_1 = \left\{ t \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} : t \in \mathbb{R} \right\}$, $S_2 = \left\{ t \begin{bmatrix} x \\ y \\ 3 \end{bmatrix} : t \in \mathbb{R} \right\}$ and $S_3 = \left\{ t \begin{bmatrix} x \\ 0 \\ z \end{bmatrix} : t \in \mathbb{R} \right\}$. **Answer: $S_1 = \left\{ t \begin{bmatrix} 1 \\ 2 \end{bmatrix} : t \in \mathbb{R} \right\}$ is line $y = 2x$,** **$S_2 = \left\{ t \begin{bmatrix} x \\ y \end{bmatrix} : t \in \mathbb{R} \right\}$ is xy plane,** **$S_3 = \left\{ t \begin{bmatrix} x \\ 0 \end{bmatrix} : t \in \mathbb{R} \right\}$ is x axis.**
B	11	Determine whether $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear operator or not. (i) $T(x, y) = (\sqrt[3]{x}, \sqrt[3]{y})$, (ii) $T(x, y) = (x, 0)$. **Answer: (i) No, (ii) Yes**
A	12	Define a function $T : \mathbb{R}^2 \to \mathbb{R}^2$ by $T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} e^x \\ e^y \end{bmatrix}$. Determine whether the T is a linear transformation or not. **Answer: No**

Method 2 ⇝ Kernel, Range and Rank-Nullity Theorem

Examples of Method-2: Kernel, Range and Rank-Nullity Theorem

A	1	Find the standard matrix of $T : \mathbb{R}^3 \to \mathbb{R}^5$ defined by $T(x, y, z) = (x - z,\ y - x,\ x - y + z,\ z - x,\ y + x)$. **Answer:** $\begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 1 & -1 & 1 \\ -1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$
A	2	Define the linear operator $T : \mathbb{R}^3 \to \mathbb{R}^3$ by $T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} x \\ -y \\ z \end{bmatrix}$. Find [T] and $T\left(\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}\right)$. **Answer:** $[T] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, T\left(\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$
B	3	If $T : \mathbb{P}_2 \to \mathbb{P}_3$ is a linear transformation defined by $T(p(x)) = x \cdot p(x + 1)$ then find the standard matrix of T. **Answer:** $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{bmatrix}$
C	4	If $T: \mathbb{P}_3 \to \mathbb{P}_3$ is a linear transformation defined by $T(p(x)) = x^2 p''(x) - 2xp'(x) + p(x + 1)$ then find the standard matrix of T. **Answer:** $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -1 & 2 & 3 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

C	5	If $T: \mathbb{M}_{2\times2} \to \mathbb{M}_{2\times2}$ is a linear transformation defined by $T(A) = MA - AM$, where $M = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$. Find the standard matrix of T relative to the standard basis. **Answer:** $\begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{2} & \mathbf{0} \\ \mathbf{-2} & \mathbf{-2} & \mathbf{0} & \mathbf{2} \\ \mathbf{0} & \mathbf{0} & \mathbf{2} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{-2} & \mathbf{0} \end{bmatrix}$
B	6	If $T: \mathbb{M}_{2\times2} \to \mathbb{M}_{2\times2}$ is a linear transformation defined by $T(A) = A - 2A^T$. Find the standard matrix of T relative to the standard basis. **Answer:** $\begin{bmatrix} \mathbf{-1} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{-2} & \mathbf{0} \\ \mathbf{0} & \mathbf{-2} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{-1} \end{bmatrix}$
B	7	Let $T: \mathbb{R}^3 \to \mathbb{R}^3$ be the linear transformation defined by $T(x, y, z) = (x + y + 2z, 2x - y + z, 3x + 3z)$ then find the bases of ker(T) and R(T). **Answer:** **Basis of ker(T)** $= \{(\mathbf{1}, \mathbf{1}, \mathbf{-1})\}$, **Basis of R(T)** $= \{(\mathbf{1}, \mathbf{2}, \mathbf{3}), (\mathbf{1}, \mathbf{-1}, \mathbf{0})\}$.
B	8	Let $T: \mathbb{R}^4 \to \mathbb{R}^3$ be the linear transformation defined by $T(x, y, z, w) = (4x + y - 2z - 3w, 2x + y + z - 4w, 6x - 9z + 9w)$ then find the bases of ker(T) and R(T). **Answer:** **Basis of ker(T)** $= \left\{\left(\frac{\mathbf{3}}{\mathbf{2}}, \mathbf{-4}, \mathbf{1}, \mathbf{0}\right)\right\}$, **Basis of R(T)** $= \{(\mathbf{4}, \mathbf{2}, \mathbf{6}), (\mathbf{1}, \mathbf{1}, \mathbf{0}), (\mathbf{-3}, \mathbf{-4}, \mathbf{9})\}$.

B	9	Let $T : \mathbb{P}_2 \to \mathbb{R}^2$ be the linear transformation defined by $T(a_0 + a_1x + a_2x^2) = (a_0 - a_1, a_1 + a_2)$ then find the bases of ker(T) and R(T). **Answer: Basis of ker(T) $= \{x^2 - x - 1\}$,** **Basis of R(T) $= \{(1, 0), (-1, 1)\}$.**
B	10	Let $T : \mathbb{P}_2 \to \mathbb{R}^2$ be the linear transformation defined by $T(p(x)) = (p(1), p'(1))$ then find the bases of ker(T) and R(T). **Answer: Basis of ker(T) $= \{x^2 - 2x + 1\}$,** **Basis of R(T) $= \{(1, 0), (0, 1)\}$.**
C	11	Let W be the vector space of all symmetric 2×2 matrices and Let $T : W \to \mathbb{P}_2$ be the linear transformation defined by $T\begin{bmatrix} a & b \\ b & c \end{bmatrix} = (a - b) + (b - c)x + (c - a)x^2$ find the bases of ker(T) and R(T). Also find rank(T) and nullity(T). **Answer: Basis of ker(T) $= \left\{\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\right\}$.** **Basis of R(T) $= \{1 - x^2, -1 + x\}$.** **rank(T) $= 2$, nullity(T) $= 1$.**
C	12	Let $T : \mathbb{M}_{2\times2} \to \mathbb{M}_{2\times2}$ be the linear transformation defined by $T(A) = AB - BA$ where, $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ then find the bases of ker(T) and R(T). Also find rank(T) and nullity(T). **Answer: Basis of ker(T) $= \left\{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right\}$,** **Basis of R(T) $= \left\{\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right\}$.** **rank(T) $= 2$, nullity(T) $= 2$**

A	13	Let $T: \mathbb{R}^3 \to \mathbb{R}^4$ be a linear transformation defined by $T(x, y, z) = (x + y + z,\ x + 2y - 3z,\ 2x + 3y - 2z,\ 3x + 4y - z)$. Find the rank(T) and then find the nullity(T) from the rank-nullity theorem. **Answer: $\mathbf{rank(T) = 2,\ \ nullity(T) = 1.}$**
B	14	Verify rank-nullity theorem for the linear transformation $T : \mathbb{P}_2 \to \mathbb{P}_2$ defined by $T\big(p(x)\big) = p(x) - x \cdot p'(x)$.
B	15	Let $T : \mathbb{P}_2 \to \mathbb{P}_3$ defined by $T\big(p(x)\big) = x \cdot p(x + 1)$ be a linear transformation. Find the rank(T) and then find the nullity(T) by using rank-nullity theorem. **Answer: $\mathbf{rank(T) = 3,\ \ nullity(T) = 0.}$**

Method 3 ⇝ Matrix Representation of Linear Transformations

Examples of Method-3: Matrix representation of a Linear Transformation

A	1	Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be the linear transformation defined by $T(x, y) = (x + y, 2x - y, 3y)$. Find the matrix for the transformation T with respect to the basis $B = \{(1, 1), (1, -1)\}$ for $\mathbb{R}^2$ and $B' = \{(1, 0, 0),\ (1, 1, 0),\ (1, 1, 1)\}$ for $\mathbb{R}^3$. **Answer:** $[T]_{B',B} = \begin{bmatrix} 1 & -3 \\ -2 & 6 \\ 3 & -3 \end{bmatrix}$
C	2	Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be the linear transformation defined by $T(x_1, x_2, x_3, x_4) = (x_1 - x_2 + 2x_3 - x_4, 2x_1 + x_2 - x_3 + 3x_4, -x_1 + 2x_2 + 3x_3 + x_4)$. Find the matrix for T with respect to the bases $A = \{(1, 1, 0, 0), (0, 1, 1, 0), (0, 0, 1, 1), (1, 0, 0, 1)\}$ and $B = \{(1, 1, 1), (1, 2, 3), (1, 3, 6)\}$. **Answer:** $[T]_{B,A} = \begin{bmatrix} -8 & 8 & 1 & -15 \\ 13 & -13 & -1 & 25 \\ -5 & 6 & 1 & -10 \end{bmatrix}$
A	3	If $T : \mathbb{P}_2 \to \mathbb{P}_3$ is a linear transformation defined by $T(p(x)) = x \cdot p(2x)$ then find the matrix $[T]_{B',B}$ with respect to the bases $B = \{1,\ x,\ x^2\}$ and $B' = \{1,\ x,\ x^2,\ x^3\}$. **Answer:** $[T]_{B',B} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$

B	4	If $T: \mathbb{P}_2 \to \mathbb{P}_2$ is a linear transformation defined by $T(p(x)) = p(x-1) + x \cdot p'(x)$ then find the matrix $[T]_{C,B}$ with respect to the bases $B = \{1,\ x-1,\ x^2\}$ and $C = \{1,\ x+1,\ x^2+2x+1\}$. **Answer:** $[T]_{C,B} = \begin{bmatrix} 1 & -4 & 6 \\ 0 & 2 & -8 \\ 0 & 0 & 3 \end{bmatrix}$
B	5	Let $T: \mathbb{R}^3 \to \mathbb{R}^3$ be the linear operator defined by $T(x,y,z) = (x-y+2z, 2x+y-z, 2y+3z)$. Find the matrix representation of T with respect to (i) S_2, (ii) S_2 & S_1 where, $S_1 = \{(1,1,0),(0,1,1),(1,0,1)\}$ and $S_2 = \{(1,1,1),(1,1,0),(1,0,0)\}$. **Answer:** $[T]_{S_2} = \begin{bmatrix} 5 & 2 & 0 \\ -3 & 1 & 2 \\ 0 & -3 & -1 \end{bmatrix}$, $[T]_{S_1,S_2} = \frac{1}{2}\begin{bmatrix} -1 & 1 & 3 \\ 5 & 5 & 1 \\ 5 & -1 & -1 \end{bmatrix}$
B	6	Let $T: \mathbb{R}^3 \to \mathbb{R}^3$ be the linear operator defined by $T(x,y,z) = (2x-y+3z, x+2y-z, 3x-y+2z)$. Find $[T]_{B_1}$ and $[T]_{B_2,B_1}$ where, $B_1 = \{(1,1,1),\ (0,1,2),(1,2,2)\}$ and $B_2 = \{(2,1,1),(1,1,0),(3,2,0)\}$. **Answer:** $[T]_{B_1} = \begin{bmatrix} 8 & 13 & 11 \\ 2 & 3 & 2 \\ -4 & -8 & -5 \end{bmatrix}$, $[T]_{B_2,B_1} = \begin{bmatrix} 4 & 3 & 5 \\ 2 & -7 & 2 \\ -2 & 2 & -2 \end{bmatrix}$

C	7	Let $T : \mathbb{M}_{2\times2} \to \mathbb{M}_{2\times2}$ be the linear transformation defined by $T(A) = A\begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} + A^T$ then find the matrix of the transformation $[T]_{X,Y}$ with respect to the bases $Y = \left\{\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right\}$ and $X = \left\{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\right\}$. **Answer:** $[T]_{X,Y} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & -1 & 0 & 1 \end{bmatrix}$
B	8	If $T : \mathbb{P}_2 \to \mathbb{P}_2$ is a linear operator defined by $T(p(x)) = (x-1)\cdot p'(x)$ then find the matrix $[T]_{B',B}$ with respect to the bases $B = \{1,\ 1+x, 1+x+x^2\}$ and $B' = \{1+x^2,\ x+x^2,\ x^2\}$. **Answer:** $[T]_{B',B} = \begin{bmatrix} 0 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 4 \end{bmatrix}$
C	9	If $T: \mathbb{P}_2 \to \mathbb{P}_2$ is a linear operator defined by $T(a+bx+cx^2) = (a+2b-c) + (2a-b)x + (b+3c)x^2$ then find $[T]_{B_1,B_2}$ and $[T]_{B_2}$ where, $B_1 = \{x^2,\ x^2-x,\ x^2-x+1\}$ and $B_2 = \{1+x,\ 1-x,\ 2x^2\}$. **Answer:** $[T]_{B_1,B_2} = \begin{bmatrix} -\frac{1}{2} & -1 & \frac{1}{2} \\ -\frac{1}{2} & -2 & -\frac{5}{2} \\ \frac{3}{2} & 1 & 1 \end{bmatrix}$, $[T]_{B_2} = \begin{bmatrix} 2 & 1 & -1 \\ 1 & -2 & -1 \\ \frac{1}{2} & -\frac{1}{2} & 3 \end{bmatrix}$

C	10	Let $T:\mathbb{R}^4 \to \mathbb{R}^4$ be the linear operator defined by $T(x,y,z,w) = (x+y-z+w, 2x-y+z+2w, x+2y-2z+w, -x+y+z-w)$. Find $[T]_{B',B}$ where, $B = \{(1,0,0,0), (1,1,0,0), (1,1,1,0), (1,1,1,1)\}$ and $B' = \{(0,0,0,1), (0,0,1,1), (0,1,1,1), (1,1,1,1)\}$. **Answer:** $[T]_{B',B} = \begin{bmatrix} -2 & -3 & 0 & -2 \\ -1 & 2 & -1 & -2 \\ 1 & -1 & 1 & 2 \\ 1 & 2 & 1 & 2 \end{bmatrix}$
B	11	If $T: \mathbb{P}_2 \to \mathbb{P}_2$ is a linear transformation defined by $T(p(x)) = p''(x) - 2x \cdot p'(x) + 3p(x)$ then find $[T]_{B_1}$ and $[T]_{B_2}$ where, $B_1 = \{1-x^2, 1+x^2, x-x^2\}$ and $B_2 = \{1, 1+2x, 1+2x+3x^2\}$. **Answer:** $[T]_{B_1} = \begin{bmatrix} 0 & 3 & -2 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $[T]_{B_2} = \begin{bmatrix} 3 & 2 & 8 \\ 0 & 1 & 2 \\ 0 & 0 & -1 \end{bmatrix}$
A	12	Let $T: \mathbb{M}_{2\times 2} \to \mathbb{R}^2$ be the linear transformation defined by $T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+d, b-c)$ then find the matrix of the transformation $[T]_{A,B}$ with respect to the bases $A = \{(2,0), (1,1)\}$ & $B = \left\{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\right\}$. **Answer:** $[T]_{A,B} = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$

Method 4 ⇝ Change of Basis

Examples of Method-4: Change of Basis

A	1	Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear operator defined by $T(x, y) = (x - 2y, -y)$. If $B_1 = \{(1, 0), (0, 1)\}$ and $B_2 = \{(2, 1), (-3, 4)\}$ then find the matrix of T (i) with respect to the basis B_1. (ii) with respect to the basis B_2 by using transition matrix P. **Answer:** $[T]_{B_1} = \begin{bmatrix} 1 & -2 \\ 0 & -1 \end{bmatrix}$, $[T]_{B_2} = \begin{bmatrix} -\frac{3}{11} & -\frac{56}{11} \\ -\frac{2}{11} & \frac{3}{11} \end{bmatrix}$
C	2	Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ is defined by $T(x, y, z) = (x + y + 2z, 2x - y + z, 3x + 3z)$. $B_1 = \{(1, 1, 1), (0, 1, 2), (1, 2, 2)\}$ and $B_2 = \{(2, 1, 1), (1, 1, 0), (3, 2, 0)\}$. (i) Find the matrix of T with respect to the basis B_1. (ii) Find the matrix of T with respect to the basis B_2 by using transition matrix P. **Answer:** $[T]_{B_1} = \begin{bmatrix} 10 & 14 & 19 \\ 4 & 5 & 7 \\ -6 & -9 & -12 \end{bmatrix}$, $[T]_{B_2} = \begin{bmatrix} 9 & 3 & 9 \\ 11 & 2 & 11 \\ -8 & -2 & -8 \end{bmatrix}$
A	3	Find the transition matrix from S_2 to S_1 where $S_2 = \{(1, 1, 0), (0, 1, 1), (1, 3, 1)\}$ and $S_1 = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$. **Answer:** $P = \begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}$
A	4	Find the transition matrix from S_2 to S_1 where $S_2 = \{(2, 1, 0), (0, 1, 2), (1, 3, 1)\}$ and $S_1 = \{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$. **Answer:** $P = \frac{1}{2}\begin{bmatrix} 3 & -1 & 3 \\ -1 & 3 & 3 \\ 1 & 1 & -1 \end{bmatrix}$

B	5	Find the transition matrix from B_1 to B_2 where $B_2 = \{(1,0,0,0),(1,1,0,0),(1,1,1,0),(1,1,1,1)\}$ and $B_1 = \{(2,1,-1,0),(0,2,1,3),(1,0,0,4),(-2,3,1,1)\}$. **Answer:** $\mathbf{P} = \begin{bmatrix} \mathbf{1} & \mathbf{-2} & \mathbf{1} & \mathbf{-5} \\ \mathbf{2} & \mathbf{1} & \mathbf{0} & \mathbf{2} \\ \mathbf{-1} & \mathbf{-2} & \mathbf{-4} & \mathbf{0} \\ \mathbf{0} & \mathbf{3} & \mathbf{4} & \mathbf{1} \end{bmatrix}$
B	6	Let $S_1 = \{v_1, v_2, v_3\}$ be a basis of a vector space V and let $T: V \to V$ be a linear operator such that $[T]_{S_1} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$. Find $[T]_{S_2}$ where $S_2 = \{w_1, w_2, w_3\}$ is the basis for V defined by $w_1 = v_1 + v_2$, $w_2 = v_2 + v_3$, $w_3 = v_3$. **Answer:** $[T]_{S_2} = \begin{bmatrix} \mathbf{2} & \mathbf{1} & \mathbf{1} \\ \mathbf{-1} & \mathbf{0} & \mathbf{-1} \\ \mathbf{2} & \mathbf{-1} & \mathbf{0} \end{bmatrix}$
B	7	Let $S_1 = \{v_1, v_2, v_3\}$ be a basis of a vector space V and let $T: V \to V$ be a linear operator such that $[T]_{S_1} = \begin{bmatrix} 3 & -1 & 2 \\ 1 & 4 & -3 \\ -2 & 0 & 1 \end{bmatrix}$. Find $[T]_{S_2}$ where $S_2 = \{w_1, w_2, w_3\}$ is the basis for V defined by $w_1 = v_1 + 2v_2$, $w_2 = 2v_1 + 5v_2$, $w_3 = v_3$. **Answer:** $[T]_{S_2} = \begin{bmatrix} \mathbf{-13} & \mathbf{-39} & \mathbf{16} \\ \mathbf{7} & \mathbf{20} & \mathbf{-7} \\ \mathbf{-2} & \mathbf{-4} & \mathbf{1} \end{bmatrix}$

C	8	Let $S_1 = \{v_1, v_2, v_3\}$ and $S_2 = \{w_1, w_2, w_3\}$ be the bases for a vector space V and let $P = \begin{bmatrix} 4 & 3 & 2 \\ 5 & -1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$ be the transition matrix from S_1 to S_2. Express w_1, w_2, w_3 as linear combinations of v_1, v_2, v_3. **Answer:** $\mathbf{w_1 = -v_1 - 5v_2 + 10v_3}$, $\mathbf{w_2 = v_1 + 4v_2 - 8v_3}$, $\mathbf{w_3 = 2v_1 + 10v_2 - 19v_3}$
C	9	Let $S_1 = \{v_1, v_2, v_3\}$ and $S_2 = \{w_1, w_2, w_3\}$ be the bases for a vector space V and let $P = \begin{bmatrix} 3 & 5 & 1 \\ 1 & 2 & 1 \\ 2 & 6 & 7 \end{bmatrix}$ be the transition matrix from S_2 to S_1. Express v_1, v_2, v_3 as linear combinations of w_1, w_2, w_3. **Answer:** $\mathbf{v_1 = 8w_1 - 5w_2 + 2w_3}$, $\mathbf{v_2 = -29w_1 + 19w_2 - 8w_3}$, $\mathbf{v_3 = 3w_1 - 2w_2 + w_3}$.
B	10	Let $\mathbb{P}_1 \to \mathbb{P}_1$ be a linear transformation defined by $T(a_0 + a_1x) = (a_0 + a_1) + a_1x$. Suppose $S_1 = \{6 + 3x, 10 + 2x\}$ and $S_2 = \{2, 3 + 2x\}$ are bases of $\mathbb{P}_1$. Find the matrix of T with respect to the basis S_1 and matrix of T with respect to the basis S_2 by using $[T]_{S_1}$. **Answer:** $[T]_{S_1} = \begin{bmatrix} \frac{2}{3} & -\frac{2}{9} \\ \frac{1}{2} & \frac{4}{3} \end{bmatrix}$, $[T]_{S_2} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$
B	11	Let $\mathbb{P}_1 \to \mathbb{P}_1$ be a linear transformation defined by $T(a + bx) = (a + 2b) + (2a - b)x$. Suppose $S_1 = \{2, 2 - x\}$ and $S_2 = \{9, x - 4\}$ are bases of $\mathbb{P}_1$. Find the matrix of T with respect to the basis S_1 and matrix of T with respect to the basis S_2 by using $[T]_{S_1}$. **Answer:** $[T]_{S_1} = \begin{bmatrix} 5 & 5 \\ -4 & -5 \end{bmatrix}$, $[T]_{S_2} = \begin{bmatrix} 9 & -\frac{38}{9} \\ 18 & -9 \end{bmatrix}$

C	12	Let $\mathbb{P}_2 \to \mathbb{P}_2$ be a linear transformation defined by $T(a + bx + cx^2) = (a + b) + (b - c)x + (a + 2c)x^2$. Suppose $S_1 = \{1, x, x^2\}$ and $S_2 = \{1, 1 + x, 1 + x + x^2\}$ are bases of $\mathbb{P}_2$. Find the matrix of T with respect to the basis S_1 and matrix of T with respect to the basis S_2 by using transition matrix P from S_2 to S_1. **Answer:** $[T]_{S_1} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 2 \end{bmatrix}$, $[T]_{S_2} = \begin{bmatrix} 1 & 1 & 2 \\ -1 & 0 & -3 \\ 1 & 1 & 3 \end{bmatrix}$
C	13	Let V be the vector space of all 2×2 symmetric matrices. Let $T: V \to V$ be the linear operator defined by $T\left(\begin{bmatrix} a & b \\ b & c \end{bmatrix}\right) = \begin{bmatrix} a + c & 2a + b \\ 2a + b & b - c \end{bmatrix}$. Suppose S_1 is the standard basis of V and $S_2 = \left\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\right\}$ then find $[T]_{S_2}$ by using transition matrix P from S_2 to S_1. **Answer:** $[T]_{S_2} = \begin{bmatrix} -1 & -2 & -1 \\ 2 & 2 & 3 \\ 0 & 1 & 0 \end{bmatrix}$
C	14	Let V be the vector space of all 2×2 matrices whose trace is zero. Let $T: V \to V$ be the linear operator defined by $T\left(\begin{bmatrix} a & b \\ c & -a \end{bmatrix}\right) = \begin{bmatrix} a - c & b + c \\ b - c & c - a \end{bmatrix}$. Suppose $S_1 = \left\{\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right\}$ and $S_2 = \left\{\begin{bmatrix} 4 & 0 \\ 0 & -4 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}, \begin{bmatrix} -2 & 1 \\ 0 & 2 \end{bmatrix}\right\}$ then find $[T]_{S_2}$ by using $[T]_{S_1}$. **Answer:** $[T]_{S_2} = \begin{bmatrix} 1 & \frac{3}{4} & -\frac{1}{4} \\ 0 & -1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

Method 5 ⇝ Similarity of Matrices

Examples of Method-5: Similarity of Matrices

A	1	Prove that every square matrix is similar to itself.
A	2	Prove that if A is similar to B, then A^2 is similar to B^2.
A	3	Prove that if A is similar to B, then A^T is similar to B^T.
A	4	Prove that if A is similar to B, then A^{-1} is similar to B^{-1}.
B	5	Show that the matrices $A = \begin{bmatrix} 120 & 45 & 75 \\ 10 & -50 & 90 \\ 60 & 30 & 80 \end{bmatrix}$ and $B = \begin{bmatrix} 80 & 15 & 95 \\ 40 & 120 & 65 \\ 70 & 55 & -50 \end{bmatrix}$ are not similar.
B	6	Show that the matrices $A = \begin{bmatrix} -125 & 432 & 85 \\ 92 & -54 & 310 \\ 198 & 76 & 215 \end{bmatrix}$ and $B = \begin{bmatrix} 198 & 76 & 215 \\ 92 & -54 & 310 \\ -125 & 432 & 85 \end{bmatrix}$ are not similar.
B	7	Show that the matrices $A = \begin{bmatrix} 10 & 20 & 30 \\ 20 & 40 & 60 \\ 30 & 60 & 90 \end{bmatrix}$ and $B = \begin{bmatrix} 40 & 15 & 20 \\ 25 & 50 & 30 \\ 65 & 65 & 50 \end{bmatrix}$ are not similar.
B	8	Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear operator defined by $T(x, y, z) = (2x - y - z, x - y, y + z)$. Suppose S_1 is the standard basis for $\mathbb{R}^3$ and $S_2 = \{(1, 2, 3), (1, 1, 0), (1, 1, 1)\}$. Verify that $\rho([T]_{S_1}) = \rho([T]_{S_2})$.
C	9	Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear operator defined by $T(x, y, z) = (x - y + 3z,\ 4y - z,\ x + y + 2z)$. Verify $\eta([T]_{S_1}) = \eta([T]_{S_2})$, Where $S_1 = \{(0, 3, 6), (6, 2, 0), (1, 5, 1)\}$ and $S_2 = \{(1, 5, 2), (4, 1, 9), (1, 4, 1)\}$.

B	10	Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear operator defined by $T(x, y, z) = (y - 3z,\ x + 9y,\ 2x - y - z)$. Verify $\rho([T]_{S_1}) = \rho([T]_{S_2})$, Where $S_1 = \{(0, 2, 5), (2, 1, 0), (1, 3, 1)\}$ and $S_2 = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$.

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